

Food Delivery Customer & Sales Analysis

Project Overview

This project analyzes a Food Delivery dataset obtained from Kaggle based on a fictional **Foodpanda** business. The objective was to examine customer purchasing behavior, revenue performance, delivery efficiency, payment preferences, customer retention, and restaurant performance. An interactive dashboard was created in Microsoft Excel using PivotTables, Pivot Charts, KPI cards, slicers, and various chart types to transform raw transactional data into meaningful business insights.

Objectives

The project aims to:

- Analyze overall sales performance.
 - Identify the highest-performing cities, restaurants, and food categories.
 - Examine customer payment preferences.
 - Monitor customer churn.
 - Evaluate delivery performance.
 - Track monthly revenue trends.
 - Generate business recommendations to improve customer retention and sales.
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Dataset Information

- **Source:** Kaggle
- **Company:** Foodpanda (Dataset)
- **Total Records:** 6,000
- **Analysis Tool:** Microsoft Excel

The dataset contains customer, restaurant, order, payment, delivery, and sales information covering three years of transactions.

Link of the data: <https://www.kaggle.com/datasets/nabihazahid/foodpanda-analysis-dataset-2025>

Data Cleaning & Preparation

The dataset was prepared in Microsoft Excel before creating the dashboard.

- The **Remove Duplicates** feature (**Data** → **Remove Duplicates**) was used to identify duplicate records, and no duplicate values were found.
- Using **Go to Special** (**Ctrl + G** → **Special** → **Blanks**), no blank cells were found in the dataset.
- Data types and formatting were reviewed to ensure consistency across all columns.
- A **order_month_year** column was created from the **order_date** field to support monthly revenue trend analysis and enable chronological visualization in the line chart.
- Multiple PivotTables were created to summarize the data, which were then used to build the dashboard visualizations and KPI cards.

Dashboard KPIs

KPI	Value
Total Revenue	4,803,149
Average Order Amount	801
Total Orders	6,000
Total Customers	6,000
Average Rating	3.0
Average Orders per Customer	25

Dashboard Analysis

Revenue by City

Multan generated the highest revenue (**1,009,004**), followed by Peshawar and Lahore. Karachi recorded the lowest revenue among the five cities. This indicates regional differences in customer demand and suggests opportunities to strengthen marketing activities in lower-performing cities.

Revenue by Category

Italian cuisine generated the highest revenue (**1,005,523**), while Dessert produced the lowest revenue (**889,128**). The popularity of Italian food indicates strong customer demand and provides opportunities for promotional campaigns focused on this category.

Revenue by Restaurant

Subway generated the highest revenue (**1,009,983**), followed closely by KFC and Pizza Hut. McDonald's recorded the lowest revenue among the listed restaurants.

Revenue by Payment Method

Customer spending is distributed almost evenly across all payment methods.

- Cash: **1,647,244 (34%)**
- Card: **1,592,170 (33%)**
- Wallet: **1,563,735 (33%)**

The balanced distribution indicates that customers are comfortable using multiple payment options, reducing reliance on a single payment channel.

Customer Churn

Out of 6,000 customers:

- **3,016 (50%)** are Active.
- **2,984 (50%)** are Inactive.

The nearly equal distribution suggests that customer retention remains an important area for improvement.

Delivery Status

Among the 6,000 recorded orders:

- Delivered: **2,060**
- Delayed: **1,972**

- Cancelled: **1,968**

The dataset records whether an order was delayed or cancelled; however, it does not include the reasons behind these outcomes. Adding fields such as **Delay Reason** and **Cancellation Reason** (for example, restaurant unavailable, payment failure, customer cancellation, rider shortage, traffic, or adverse weather) would help identify recurring operational issues and support targeted improvements.

Monthly Revenue Trend

Revenue remained relatively stable throughout most of the analysis period, with noticeable fluctuations across different months. Revenue peaked during **March 2025**, while the final months of 2025 experienced a gradual decline. Monitoring seasonal trends can help improve demand forecasting and promotional planning.

Key Findings

- Total revenue reached **4.8 million**.
 - Italian cuisine generated the highest revenue.
 - Subway was the highest-performing restaurant.
 - Multan recorded the highest city revenue.
 - Customer payments were almost evenly distributed among Cash, Card, and Wallet.
 - Active and inactive customers were nearly equal in number.
 - Delivered, delayed, and cancelled orders occurred in almost equal proportions.
 - The average order amount was **801**.
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Business Recommendations

Improve New Customer Retention

Encourage newly registered customers to place their first order by providing welcome offers for the first 7–10 days such as discount vouchers, free delivery for the first 1-2 order, or limited-time promotional codes. Early engagement can reduce the number of inactive accounts and increase customer retention.

Introduce Loyalty Programs

Reward long-term customers with exclusive discounts, or free delivery. Periodically recognizing top customers with special offers can encourage repeat purchases and strengthen customer loyalty. Make best use of the loyalty points.

Encourage Higher Spending

Introduce threshold-based promotions such as:

Spend above 801 and receive 10% off - Since 801 is the current average order amount, this strategy may increase average order value.

Promote High-Performing Categories

Develop marketing campaigns, combo meals, and promotional bundles featuring Italian cuisine to capitalize on its strong sales performance.

Optimize Regional Marketing

Maintain strong marketing efforts in high-performing cities while increasing promotional activities in lower-performing cities, particularly Karachi, to improve regional sales performance.

Improve Delivery Operations

Extend the dataset by recording cancellation and delay reasons. Analyzing these factors will help reduce operational inefficiencies, improve delivery performance, and increase customer satisfaction.

Evaluate Restaurant Performance

Monitor customer ratings and reviews for partner restaurants. Restaurants receiving consistently poor ratings should be encouraged to improve food quality and service standards.

Payment Method Promotions

Since payment methods contribute almost equally to revenue, promotional campaigns such as cashback offers, bank partnerships, or digital wallet discounts can be used across all payment methods to encourage additional purchases.

Conclusion

The Food Delivery Customer & Sales Analysis dashboard provides a comprehensive overview of sales performance, customer behavior, delivery operations, and restaurant performance. By leveraging Excel's analytical capabilities, the dashboard transforms transactional data into actionable business insights. The findings highlight opportunities to strengthen customer retention, improve operational efficiency, increase average order value, and optimize marketing strategies. Implementing the recommended initiatives can help improve customer satisfaction while supporting long-term business growth.